

Haleh Damirchi

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Machine learning researcher and Ph.D. graduate from Queen's University with seven years of experience developing deep learning models in Python and PyTorch, including four years in probabilistic and time-series forecasting.

Work Experience

Graduate Research Fellow

May 2021 – April 2026

Queen's University

AIIM Lab, RCV Lab

- Co-developed a multitask counting and localization network using pretrained VGG and point-level supervision.
- Enhanced efficiency and accuracy of FPV Pedestrian path prediction by designing a transformer-based context-aware multimodal trajectory forecasting model.
- Improved the accuracy of pedestrian path prediction by designing a socially informed CVAE-based forecaster using online pseudo-trajectory augmentation for challenging training samples.
- Boosted the rigor of model comparisons by conducting an analysis of evaluation methods for trajectory forecasting.

University Lecturer

Jan 2026 – April 2026

Queen's University

Introduction to Data Science

- Taught SQL and relational databases, along with data analysis and visualization using Pandas and Matplotlib.
- Covered machine learning topics including dimensionality reduction, imputation, noise reduction, k-fold cross-validation, logistic regression, and classification.

Technical Skills

Programming & Tools: Python, C, SQL, MATLAB, Bash, SLURM, HPC clusters, Git, L^AT_EX, MySQL, FastAPI

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn

Architectures and Methods: CNNs, RNNs, Transformers, probabilistic forecasting, multimodal learning

Selected Publications

- [1] **H. Damirchi**, M. Greenspan and A. Etemad, "A Comprehensive and Critical Analysis of Metrics and Evaluation Practices in Pedestrian Trajectory Forecasting", Under Review.
- [2] **H. Damirchi**, A. Etemad and M. Greenspan, "Socially-informed Reconstruction for Pedestrian Trajectory Forecasting", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).
- [3] **H. Damirchi**, M. Greenspan and A. Etemad, "Context-Aware Pedestrian Trajectory Prediction with Multimodal Transformer", IEEE International Conference on Image Processing (ICIP 2023).
- [4] M. Zand, **H. Damirchi**, A. Farley and M. Molahasani, "Multiscale Crowd Counting and Localization by Multitask Point Supervision", IEEE International Conference on Acoustics (ICASSP 2022).

Education

Queen's University, Ph.D. in Electrical and Computer Engineering

2026

Amirkabir University of Technology, M.Sc. in Electrical Engineering

2020

University of Tabriz, B.Sc. in Electrical Engineering

2016

Selected Projects

SlouchLess | *Python*

- Packaged and published standalone Windows and Linux releases for an application that detects prolonged slouching from webcam video and provides real-time posture reminders.
- Built a data collection and training pipeline for learning posture patterns.

PaperScraper | *Python*

- Built a retrieval-augmented LLM pipeline to extract datasets, sample counts, and evaluation metrics from research papers, with citations to the supporting source text.
- Implemented PDF parsing and OCR, HyDE retrieval with embeddings, and schema-constrained LLM outputs for structured information extraction.
- Developed an automated pipeline to evaluate extracted outputs against ground truth using statistical similarity metrics and LLM-as-a-judge scoring.

Speaker Extraction | *TensorFlow, Pytorch, Matlab*

- Enhanced target speaker extraction performance by developing stacked BLSTM models that outperformed DNN and MSE baselines across PESQ, STOI, and SDR.
- Improved speech quality and intelligibility by designing and implementing psychoacoustic loss functions using Mel and Gammatone filter banks and perceptual hearing models.